

$$= 0.2 \text{ eV}$$

30

C

The ground state hydrogen atoms have electrons starting at E_1 .
For ejected electron with highest speed, requires photon of lowest wavelength.

$$\frac{hc}{\lambda} = \text{ionisation energy} + KE_{max}$$

$$\frac{(6.63 \times 10^{-34})(3 \times 10^8)}{7.5 \times 10^{-9}} = 13.60(1.6 \times 10^{-19}) + \frac{1}{2}(9.11 \times 10^{-31})v^2$$

$$v = 7.31 \times 10^6 \text{ m s}^{-1}$$